

Construction Robotics Competency Regimes

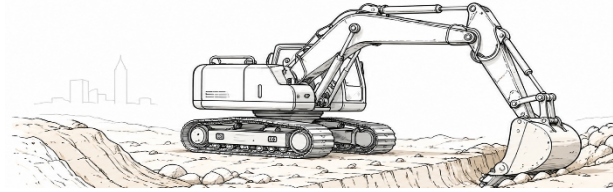
Five dominant failure-physics evidence domains

1



Earthwork and Material Processing

Evolving soil resistance, terrain interaction, and material heterogeneity



48.0%

n=36

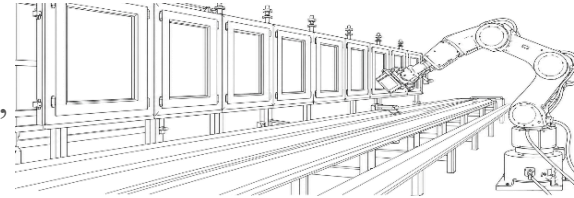


2



Structural Assembly and Installation

Contact jamming, tolerance amplification, and insertion sensitivity



24.0%

n=18

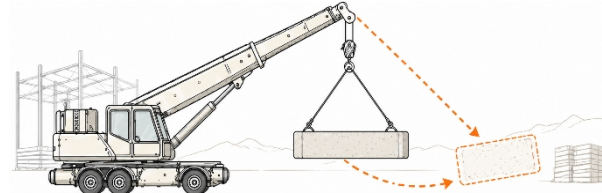


3



Material Placement and Lifting

Suspended-load dynamics and placement uncertainty



10.7%

n=8

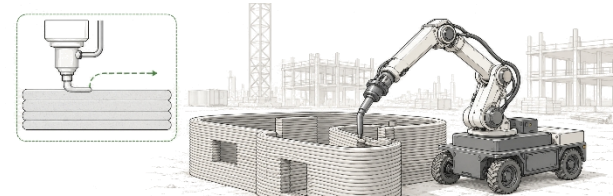


4



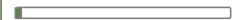
Additive Manufacturing and Surface Processing

Cumulative process-state drift and deposition or coverage consistency



2.7%

n=2

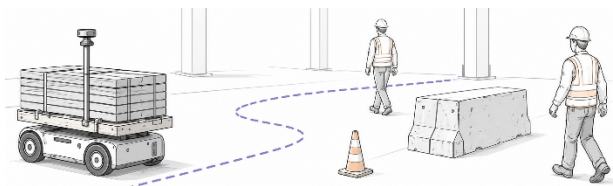


5



Navigation, Layout, and Logistics Support

Mixed-agent flow, occlusion, path uncertainty, and human co-presence



14.7%

n=11

